

In 1911 Ernest Rutherford put forward the planetary model of the atom, consisting of the nucleus surrounded by orbiting electrons, just like the planets orbit the Sun in our solar system. Rutherford's model made a number of significant improvements over the preceding plum-pudding model but was unable to explain the well-known phenomenon of each element emitting a characteristically unique set of radiation (spectra) when heated. To account for this, a new model of the atom needed to be developed. This chapter outlines this new model of the atom—one that can explain the behaviour of elements and the types of compounds that they create.

Science understanding

- atoms can be modelled as a nucleus, surrounded by electrons in distinct energy levels, held together by electrostatic forces of attraction between the nucleus and electrons; the location of electrons within atoms can be represented using electron configurations
- flame tests and atomic absorption spectroscopy (AAS) are analytical techniques that can be used to identify elements; these methods rely on electron transfer between atomic energy levels and are shown by line spectra
- the structure of the periodic table is based on the atomic number and the properties of the elements
- the elements of the periodic table show trends across periods and down main groups, including in atomic radii, valencies, 1st ionisation energy and electronegativity as exemplified by groups 1, 2, 13–18 and period 3

3.1 Electronic structure of atoms



FIGURE 3.1.1 The spectacular colours in this Australia Day fireworks display are emitted by metal atoms that have been heated to very high temperatures.

When fireworks explode, they create a spectacular show of coloured lights (Figure 3.1.1). The light is produced by metal atoms that have been heated by the explosion. These coloured lights posed a significant problem for early scientists. The planetary model for the atom put forward by Rutherford could not explain why each metal produced a colour of light unique to that particular metal. However, the light was a clue that ultimately led to a better understanding of the arrangement of electrons in atoms.

BOHR MODEL OF THE ATOM

In 1913, Niels Bohr developed a new model of the hydrogen atom that explained the production of the light characteristic of hydrogen. The **Bohr model** proposed the following.

- Electrons revolve around the nucleus in fixed, circular orbits.
- The electrons' orbits correspond to specific energy levels or shells in the atom.
- Electrons can only occupy these fixed energy levels and cannot exist between two energy levels.
- Electron orbits of larger radii correspond to energy levels of higher energy.

In the Bohr model, the electron orbits the nucleus at a fixed distance from the nucleus. This region of space the electron occupies is known as an **electron shell**. The shell corresponds with a fixed quantity of energy and any electrons occupying that shell possess the same quantity of energy, hence the shell is also referred to as an **energy level**.

Bohr also suggested that it is possible for electrons to move between the energy levels by absorbing or emitting energy in the form of light. Bohr's model (Figure 3.1.2) gave close agreement between the calculated theoretical frequencies, or colours, of light emitted by hydrogen atoms and the experimentally observed values.

Scientists quickly extended Bohr's model of the hydrogen atom to other atoms. Scientists proposed that electrons were grouped in different energy levels. The lowest energy levels contained electrons closest to the nucleus (Figure 3.1.3).

The orbit in which an electron moved depended on the energy of the electron; electrons with low energy are in orbits closest to the nucleus while high-energy electrons are in outer orbits.

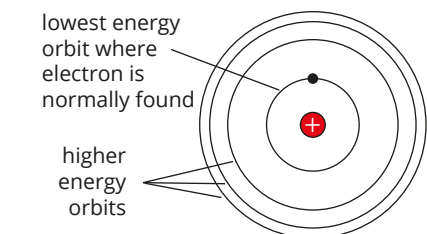


FIGURE 3.1.2 The Bohr model of a hydrogen atom. Bohr suggested that electrons moved in orbits of particular energies.

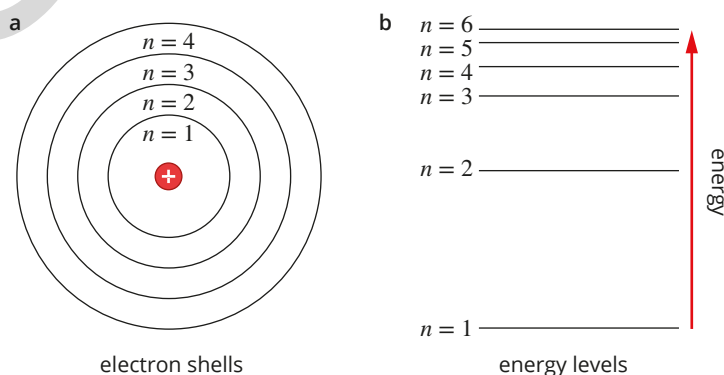


FIGURE 3.1.3 (a) The electron shells of an atom (n) are labelled using integers. The first shell is the shell closest to the nucleus and the radius of each shell increases as the shell number increases. (b) Each shell corresponds to an energy level that electrons can occupy. The first shell has the lowest energy and the energy increases as the shell number increases.

The lowest energy shell is the shell closest to the nucleus and is labelled $n = 1$. Shells with higher values of n correspond to higher energy levels. As the values of n increase, the energy levels get closer together.

The outermost shell of an atom is known as the atom's **valence shell**. The electrons in this outer shell are called **valence electrons**. These electrons require the least amount of energy to be removed from the atom. The valence electrons are involved in chemical reactions. Consequently, if you know the number of valence electrons in the atoms of an element, then you can predict the chemical properties of the element.

Later in this chapter, and in Chapters 4–6 you will learn about how atoms tend to lose, gain or share valence electrons in order to achieve eight electrons in their outer shell when they are involved in chemical reactions. This is known as the **octet rule**.

The Bohr model of the atom was revolutionary when it was proposed. Before Bohr, Rutherford and other scientists believed that electrons could orbit the nucleus at any distance from the nucleus. This picture of the electrons was based on how scientists observed the world around them. For example, planets can revolve at varying distances around the Sun. However, Bohr's theory stated that electrons only occupy specific, circular orbits. This was the first suggestion that the physics inside atoms might be very different from the physics we experience in our daily lives.

The shell model of the atom that developed from the work of Niels Bohr could explain why light was emitted when atoms were heated, however, there were some things that the model could not explain. Despite these limitations, the Bohr model is still a valuable predictive tool for the behaviour of some electrons.

EXTENSION

Quantum mechanical model of the atom

The word *quantum* means 'a specific amount'. In the Bohr model, the electrons can only have specific amounts of energy depending on which shell they are in. The energy of the electrons is said to be quantised.

In 1926 the German scientist Erwin Schrödinger proposed that electrons behaved as waves around the nucleus. This was fundamentally different from Bohr who treated electrons as tiny, hard particles.

Using a mathematical approach and his wave theory, Schrödinger developed a model of the atom called quantum mechanics. This is the model of the atom that most scientists use today.

The equations used in quantum mechanics are complex. The electron is now seen as a vague, elusive object that behaves like a cloud of negative charge

(Figure 3.1.4). Unlike the Bohr model, there are no definite orbits for electrons in this model of the atom. Electrons are thought to move in regions of space surrounding the nucleus called orbitals.

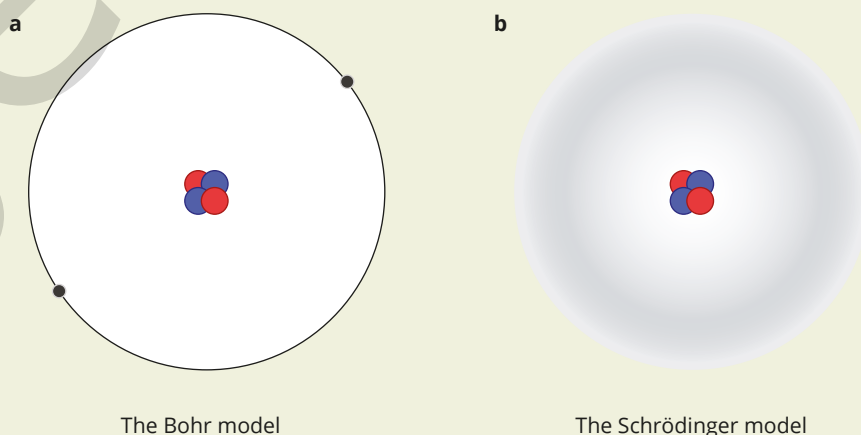


FIGURE 3.1.4 (a) Bohr regarded electrons as particles that travel along a defined path in circular orbits. (b) In Schrödinger's quantum mechanical approach, the electrons behave as waves and occupy a three-dimensional space around the nucleus. The region occupied by the electrons is known as an orbital.

3.1 Review

SUMMARY

- When atoms are heated, they emit light that is characteristic of that element.
- The Bohr model of the atom was the first atomic model to explain the origin of emission spectra.
- The Bohr model assumes that electrons can only exist in fixed, circular orbits of specific energies. These orbits later came to be known as energy levels or shells.

KEY QUESTIONS

- 1 Explain how emission spectra are evidence for electron shells in the Bohr model.
- 2 What four assumptions did Bohr make about the electronic structure of the atom?
- 3 What form of energy is emitted when an electron moves from a higher energy shell to a lower energy shell?

3.2 Electron arrangement in the periodic table

Chapter 2 looked at research into elements and the nature of atoms, the existence of subatomic particles, their charge and mass, the way these particles are arranged in an atom, and the way they behave. As scientists' understanding of the atom improved, and more elements were discovered, a way of organising this knowledge was needed.

THE PERIODIC TABLE

In the 19th century, chemists wanted to use information they had gathered about the elements to organise the elements into a useful and practical format. The work of these scientists led to what we now know as the periodic table. The early forms of the periodic table were very different from the one we use today; many elements had not been discovered and scientists only had limited information about some other elements. An early form of the periodic table was created by Russian chemist Dimitri Mendeleev (Figure 3.2.1).

At the time the early periodic table was being constructed, the existence of subatomic particles was unknown, and so the elements were placed in order of the increasing mass of the atoms. Mendeleev recognised that the chemical properties of the elements varied periodically with increasing atomic mass. He arranged elements with similar properties in vertical columns. Mendeleev proposed his periodic law—the properties of elements vary periodically with their atomic weights.

We now know that the atomic number (number of protons) is what makes one element fundamentally different from another element. The elements in the modern periodic table are therefore arranged in rows in order of increasing atomic number.

Chemists use the number of electrons in the outer shell (valence electrons) to organise the elements into columns. The form of the periodic table in common use is shown in Figure 3.2.2.

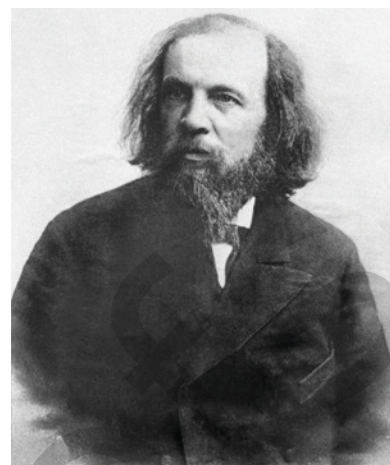


FIGURE 3.2.1 Russian chemist Dimitri Mendeleev is commonly acknowledged as the creator of the basis for the modern periodic table.

CHEMFILE

Relative atomic mass

The property of an element that Mendeleev called atomic weight is now more commonly called relative atomic mass. Relative atomic mass should not be confused with the mass number. Scientists at the time of Mendeleev did not know about subatomic particles. They only knew the relative masses of atoms of each element.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H hydrogen 1.008																	2 He helium 4.003
3 Li lithium 6.968	4 Be beryllium 9.012											5 B boron 10.82	6 C carbon 12.01	7 N nitrogen 14.01	8 O oxygen 16.00	9 F fluorine 19.00	10 Ne neon 20.18
11 Na sodium 22.99	12 Mg magnesium 24.31											13 Al aluminium 26.98	14 Si silicon 28.09	15 P phosphorus 30.97	16 S sulfur 32.07	17 Cl chlorine 35.45	18 Ar argon 39.95
19 K potassium 39.10	20 Ca calcium 40.08	21 Sc scandium 44.96	22 Ti titanium 47.88	23 V vanadium 50.94	24 Cr chromium 52.00	25 Mn manganese 54.94	26 Fe iron 55.85	27 Co cobalt 58.93	28 Ni nickel 58.69	29 Cu copper 63.55	30 Zn zinc 65.38	31 Ga gallium 69.72	32 Ge germanium 72.59	33 As arsenic 74.92	34 Se selenium 78.96	35 Br bromine 79.90	36 Kr krypton 83.80
37 Rb rubidium 85.47	38 Sr strontium 87.62	39 Y yttrium 88.91	40 Zr zirconium 91.22	41 Nb niobium 92.91	42 Mo molybdenum 95.94	43 Tc technetium	44 Ru ruthenium 101.1	45 Rh rhodium 102.9	46 Pd palladium 106.4	47 Ag silver 107.9	48 Cd cadmium 112.4	49 In indium 114.8	50 Sn tin 118.7	51 Sb antimony 121.8	52 Te tellurium 127.6	53 I iodine 126.9	54 Xe xenon 131.3
55 Cs caesium 132.9	56 Ba barium 137.3	57–71 *La lanthanum	72 Hf hafnium 178.5	73 Ta tantalum 180.9	74 W tungsten 183.9	75 Re rhenium 186.2	76 Os osmium 190.2	77 Ir iridium 192.2	78 Pt platinum 195.1	79 Au gold 197.0	80 Hg mercury 200.6	81 Tl thallium 204.4	82 Pb lead 207.2	83 Bi bismuth 209.0	84 Po polonium	85 At astatine	86 Rn radon
87 Fr francium	88 Ra radium 226.0	89–103 **Ac actinium	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium	113 Nh nihonium	114 Fl flerovium	115 Mc moscovium	116 Lv livermorium	117 Ts tennessine	118 Og oganesson
key:																	
Atomic number																	
Symbol																	
Name																	
Standard atomic weight																	
		* Lanthanide series	58 Ce cerium 140.1	59 Pr praseodymium 140.9	60 Nd neodymium 144.2	61 Pm promethium	62 Sm samarium 150.4	63 Eu europium 152.0	64 Gd gadolinium 157.3	65 Tb terbium 158.9	66 Dy dysprosium 162.5	67 Ho holmium 164.9	68 Er erbium 167.3	69 Tm thulium 168.9	70 Yb ytterbium 173.0	71 Lu lutetium 175.0	
		** Actinide series	90 Th thorium 232.0	91 Pa protactinium	92 U uranium 238.0	93 Np neptunium	94 Pu plutonium	95 Am americium	96 Cm curium	97 Bk berkelium	98 Cf californium	99 Es einsteinium	100 Fm fermium	101 Md mendelevium	102 No nobelium	103 Lr lawrencium	

FIGURE 3.2.2 This is the most common form of the periodic table in use.

CHEMFILE

Element 117 and the Canberra scientists

Four atoms of a super heavy element known as number 117 have been created in a German laboratory as part of a collaboration with Professor David Hinde (Figure 3.2.3), director of the heavy ion accelerator facility at the Department of Nuclear Physics at the Australian National University.

Because the element is radioactive, the atoms disappeared within one-tenth of a second, but the scientists were able to confirm the element's first observation in 2010. This new element was given the temporary name of ununseptium ('one-one-seven' in Latin), and in 2015 was given the name tennessine by the International Union of Pure and Applied Chemistry (IUPAC). The name recognises the contribution of a team from Oak Ridge National Laboratory in Tennessee, USA, one of the teams that confirmed its existence. The atoms of element 117 are the second-heaviest atoms ever observed, being 40% heavier than an atom of lead (element 118 is now the heaviest).

Professor Hinde said that because such an extremely small amount of the element was synthesised, it will not be practical to use it. But the Australian researchers will continue to concentrate on the quantum science behind the element's synthesis and attempt to produce another super-heavy element.



FIGURE 3.2.3 Professor David Hinde is director of the heavy ion accelerator facility at the Department of Nuclear Physics at the Australian National University in Canberra.

The modern periodic table has several key features.

- The periodic table is arranged in order of increasing atomic number.
- The atomic number is located above each chemical symbol.
- The relative atomic mass is located under each chemical symbol.
- The horizontal rows are known as **periods** and are labelled 1–7.
- The vertical columns are known as **groups** and are labelled 1–18.
- **Main group elements** are elements in groups 1, 2 and 13–18.
- The elements in groups 3–12 are known as **transition metals**.
- The lanthanoids and actinoids in periods 6 and 7 are shown separate from the main part of the table simply to make the table itself smaller and easier to manage.

Some periodic tables you will see also indicate other properties of the elements such as boiling point or whether the element is a solid, liquid or gas at room temperature.

Groups

Elements in the periodic table are arranged into vertical columns called groups. For main group elements, the group number can be used to determine the number of valence electrons (outer-shell electrons) in an atom of the element.

In groups 1 and 2, the number of valence electrons is equal to the group number. For example, magnesium is in group 2 and therefore has two valence electrons.

In groups 13–18, the number of valence electrons is equal to the group number minus 10. For example, oxygen is in group 16 so oxygen has six outer-shell electrons. Similarly, neon is in group 18 so neon has eight valence electrons. Helium is an important exception. It is located in group 18 but only has two valence electrons. Helium is placed in group 18 because it is unreactive, like other group 18 elements. This information is summarised in Table 3.2.1.

TABLE 3.2.1 The number of valence electrons in elements belonging to each group

Group	Number of valence electrons
1	1
2	2
13	3
14	4
15	5
16	6
17	7
18	8*

*Exception: helium has only two valence electrons.

The electrons in the outer shell of an atom (the valence electrons) are the electrons that are involved in chemical reactions. As a consequence, the number of valence electrons determines many of the chemical properties that an element exhibits.

Table 3.2.2 shows the names of some groups in the periodic table.

TABLE 3.2.2 Names of different groups in the periodic table

Group	Name
1	alkali metals
2	alkaline earth metals
17	halogens
18	noble gases

Because elements in the same group have the same number of valence electrons, elements in the same group have similar chemical properties. For example, the alkali metals are elements in group 1 (with the exception of hydrogen). They all have one valence electron (Figure 3.2.4) and are all relatively soft metals and are highly reactive with water and oxygen.

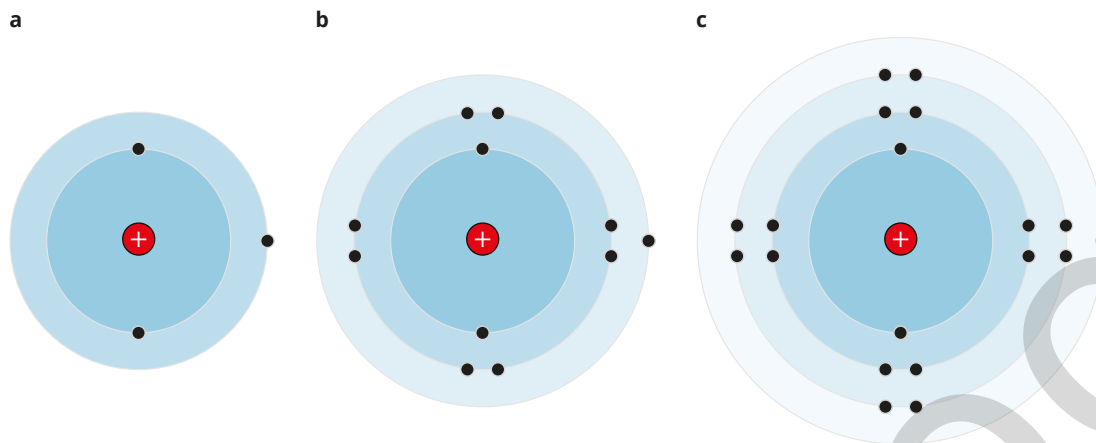


FIGURE 3.2.4 The arrangement of electrons in the first three alkali metals (a) lithium (b) sodium and (c) potassium.

Fluorine, chlorine, bromine and iodine are halogens (group 17). They all have seven valence electrons and are all coloured and highly reactive.

The **noble gases** (group 18) are a particularly interesting group. The noble gases have a very stable electron arrangement: helium has a full outer shell of two electrons and the other members of this group have a full outer shell of eight electrons. Chemical reactions involve the rearrangement of valence electrons to achieve a stable outer shell. Noble gases have a stable electron configuration, so they do not tend to lose or gain electrons. This means that the noble gases have low reactivity or are **inert**.

Periods

The horizontal rows in the periodic table are called periods. Periods are numbered 1–7. The number of a period gives information about the arrangement of electrons in each atom. The period an element is located within is equal to the number of occupied electron shells in the element's atoms. For example, the outer shell of magnesium and chlorine (Figure 3.2.5) is the third shell and both of these elements are in period 3.

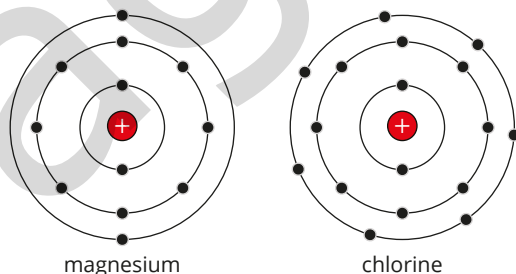


FIGURE 3.2.5 The arrangement of electrons in magnesium and chlorine. The valence electrons of both atoms occupy the third shell as they are both in period 3.

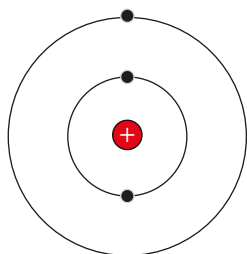


FIGURE 3.2.6 The Bohr diagram for the lithium atom with three electrons. Two electrons occupy the inner shell and the remaining one occupies the outer or valence shell.

Electron configuration

In all atoms, the electrons are as close to the nucleus as possible. This means that electrons will generally occupy inner shells before outer shells. For example, an atom of lithium has three electrons. Two electrons will occupy the first shell as that is all it can hold. The third electron is in the second shell.

A **Bohr diagram** is a simple diagram that shows the arrangement of electrons around the nucleus. In such diagrams, only the shells that are occupied are drawn. The Bohr diagram for lithium is shown in Figure 3.2.6.

A more succinct and concise way of communicating the arrangement of electrons in an atom rather than a Bohr diagram is the **electron configuration**. The electron configuration is the number of electrons in each shell separated by commas, starting at the shell with the lowest energy (closest to the nucleus).

For example, in the previous example, we saw that a lithium atom has two electrons in the inner, lowest energy shell, followed by one electron in the outer, valence shell.

Thus the electron configuration for lithium is 2,1.

To be able to determine the electron configuration for any of the first twenty elements we can simply use an element's location on the periodic table to indicate the total number of electrons present, the number of shells it has and how many electrons occupy each shell.

A maximum of two electrons can fit into the first shell, eight in the second shell and eight in the third shell. Using this information it is possible to work out the electron configurations of atoms as shown in Worked example 3.2.1.

Worked example 3.2.1

DETERMINING THE ELECTRON CONFIGURATION OF AN ATOM

Determine the electron configuration of sulfur.	
Thinking	Working
Locate sulfur (S) on the periodic table and find its atomic number. This is the number of protons in the nucleus.	16 protons
As atoms are uncharged, they have an equal number of positive charges (protons) as negative charges (electrons).	16 electrons
Start allocating electrons from lowest energy to highest, ensuring each shell is complete before progressing to the next shell. The first shell can accommodate a maximum of two electrons. There are 14 electrons remaining to be allocated.	2
The second shell can accommodate a maximum of eight electrons. There are six electrons remaining to be allocated.	2,8
The remaining six electrons (less than a complete third shell) are allocated into the third (valence) shell.	2,8,6

Worked example: Try yourself 3.2.1

DETERMINING THE ELECTRON CONFIGURATION OF AN ATOM

Determine the electron configuration of argon.

Although this is an easy pattern use to determine the electron configuration of any of the first 20 elements, this pattern does not continue from scandium onward. Scandium is the first of the transition metals, a region of the periodic table also known as the *d*-block. To determine the electron configuration of the transition metals onward, a greater understanding of the Schrödinger (quantum mechanical) model of the atom is required, which is beyond the scope of this course.

Ions

Atoms have an equal number of positive charges (protons) and negative charges (electrons). This gives them a neutral or zero overall charge. The number of electrons an atom has, however, is not fixed. Through chemical reactions or the addition of energy, the total number of electrons an atom possesses can be increased or decreased. This in turn also changes the electron configuration of the atom.

The electrons are gained to or lost from the valence shell. This shell contains the outermost and most weakly held electrons. When the number of positive charges (protons) does not equal the number of negative charges (electrons) a charged particle, known as an **ion**, is the result.

As electrons are negatively charged, if an atom obtains additional electron(s) and the number of positive charges (protons) remains constant; the atom will find itself with an overall negative charge equal to the number of electrons gained. Negatively charged ions are known as **anions**.

For example, if a chlorine atom with an electron configuration of 2,8,7 gains one valence electron, its electron configuration becomes 2,8,8. Having a total of 17 protons and now 18 electrons, the chlorine atom acquires a negative charge and becomes a chloride ion, denoted as Cl^- .

Conversely, if an atom loses electrons and the number of positive charges (protons) remains constant, the atom will have an overall positive charge equal to number of electrons lost. Positively charged ions are known as **cations**.

For example, if a sodium atom with an electron configuration of 2,8,1 loses its one valence electron, its electron configuration becomes 2,8. Having a total of 11 protons and only 10 electrons, the sodium atom now acquires a positive charge and becomes a sodium ion, denoted as Na^+ .

Table 3.2.3 shows the electron configuration of the atom and ion of some of the first 20 elements.

TABLE 3.2.3 Atom and ion electron configuration

Atom		Ion	
Symbol	Electron configuration	Symbol	Electron configuration
Li	2,1	Li^+	2
F	2,7	F^-	2,8
Na	2,8,1	Na^+	2,8
Mg	2,8,2	Mg^{2+}	2,8
Al	2,8,3	Al^{3+}	2,8
S	2,8,6	S^{2-}	2,8,8
Ca	2,8,8,2	Ca^{2+}	2,8,8

Notice that the ions share the same electron configuration as three of the noble gases—helium (2), neon (2,8) or argon (2,8,8).

The formation of ions and the associated compounds will be covered in greater depth in Chapter 5.

i In the Bohr model, the maximum number of electrons in the first shell is 2, the second shell is 8 and the third shell is 8.

i An ion is any atom of an element with more or fewer electrons than protons. If an atom loses electrons, it becomes positively charged as there are now more protons than electrons. A positively charged ion is called a cation. If an atom gains electrons, it becomes negatively charged and is called an anion.

3.2 Review

SUMMARY

- The periodic table is a tool for organising elements according to their chemical and physical properties.
- The elements of the periodic table are arranged in order of increasing atomic number.
- Columns in the periodic table are known as groups and are numbered 1–18.
- The number of valence electrons in an atom of an element can be determined by the group in which it is located (Table 3.2.1 on page 28).
- Certain groups in the periodic table have specific names (Table 3.2.2 on page 28).
- The main group elements are in groups 1, 2 and 13–18 in the periodic table.
- Elements in between the main group elements (groups 3–12) are known as transition metals.
- Rows in the periodic table are known as periods and are numbered 1–7.
- The number of occupied electron shells of an atom of an element is equal to the number of the element's period.
- Bohr diagrams can be used to show the number and location of electrons surrounding a nucleus.
- The electron configuration of an atom indicates the arrangement of electrons in each shell.
- A maximum of two electrons can fit into the first shell, eight in the second shell and eight in the third shell.

KEY QUESTIONS

- 1 Construct the electron configurations for atoms with:
 - a 5 electrons
 - b 12 electrons
 - c 20 electrons.
- 2 Write the electron configuration of the following elements.
 - a Be
 - b S
 - c Ar
 - d Mg
 - e Ne
- 3 Write the name and symbol of the element with the electron configuration:
 - a 2
 - b 2,7
 - c 2,8,3
 - d 2,5
 - e 2,8,7.

3.3 Trends in the periodic table

The periodic table does not just provide information about an element's electron configuration. It can also be used as a tool for summarising the relative properties of elements and explaining the trends observed in those properties.

You have already seen how the group number of an element determines how many valence electrons an atom of that element has and how the period indicates how many electron shells are occupied in an atom of an element. Properties such as atomic radius, ionisation energy and electronegativity show common trends in the periodic table and can be predicted and explained by examining the underlying forces present in each atom.

ELECTROSTATIC ATTRACTION

The behaviour of magnets when brought in close proximity of each other is a familiar phenomenon. If two north or south poles are brought together, they repel, whereas if a north and south pole are brought together, they attract. Indeed, the saying 'opposites attract' is a commonly used phrase and it is generally understood that positive and negative charges attract just as the north and south poles of two magnets do.

Although magnetic and electrostatic attraction can seem to be similar, the nature of the force which is present is different. The electrostatic attraction which exists between positive and negative charges is the force that in addition to holding individual atoms together, also holds atoms together in molecules and then these molecules together to in all forms of matter.

The force of attraction between two charges is described mathematically by Coulomb's law:

$$F = k \frac{q_1 q_2}{r^2}$$

where q_1 and q_2 are the magnitude of the charges involved, r is the distance between them and k is a constant. Although the use of this formula in calculations is not part of this course, understanding how changes to the quantities involved in the formula affect the strength of the electrostatic attraction, allows us to explain many trends in the periodic table.

From this equation two important points can be discerned.

- The strength of any electrostatic attraction is directly proportional to the magnitude of the charges involved. That is, as the size of the charges increases, the force of attraction between the charges also increases.
- The strength of any electrostatic attraction is inversely proportional to the distance between the charges squared. That is, as the distance between the charges increases, the force of attraction decreases. Note that as the distance, r , is squared, any change to the distance between the charges will have a more significant effect on the strength of the attraction than a change of similar magnitude to the size of the charges.

It is these two factors, the charges (on ions, electrons or protons) and the distance (between these charged particles) involved in each electrostatic attraction that allow us to explain the other trends present as we move through the periodic table.

Core charge

The **core charge** or **effective nuclear charge** of an atom is a measure of the attractive force felt by the valence electrons towards the nucleus. Core charge, although not a physically measurable quantity, is a useful concept to help explain patterns in the periodic table.

Consider an atom of lithium, which has an atomic number of three. It has three protons in its nucleus, two electrons in the first shell and one electron in the second shell (Figure 3.3.1).

- The valence electron is attracted to the three positive charges in the nucleus.
- This electron is also repelled by the two electrons in the inner shell, reducing the attraction it experiences to the nucleus.

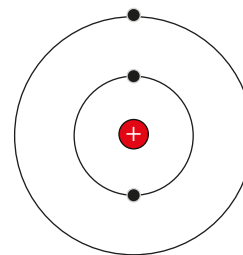


FIGURE 3.3.1 A lithium atom with one valence electron and two electrons in the inner shell. The atom has a core charge of +1.

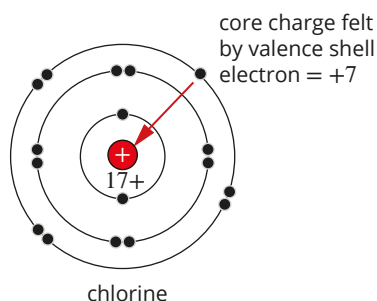


FIGURE 3.3.2 A Bohr diagram for the chlorine atom showing the valence shell electron shielded by 10 electrons in shells $n = 1$ and $n = 2$.

- The electrons in the inner shell are said to shield the valence electron from the full attraction of the nucleus.
- The valence electron is therefore effectively attracted to the nucleus as if there were a +1 nuclear charge.
- This atom is therefore said to have a core charge of +1.

In atoms with two or more shells filled with electrons, the attraction between the nucleus and valence electrons is reduced by repulsion between the inner shell electrons and the valence shell electrons.

For example, an atom of chlorine (Figure 3.3.2) has 17 protons and seven valence electrons; the number of electrons in the inner shells is 10. The core charge of a chlorine atom is therefore $17 - 10 = +7$.

i In general:
 core charge = number of protons in the nucleus – number of total inner-shell electrons

Consider the atoms of two different elements in group 1, lithium and sodium shown in Figure 3.3.3.

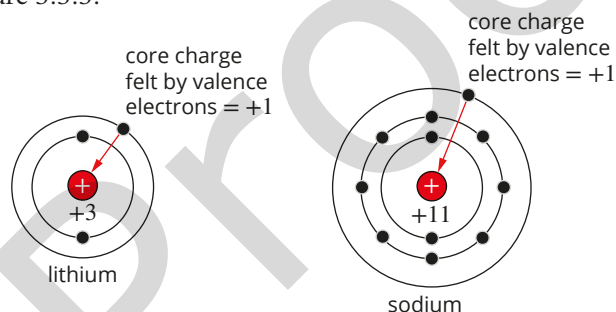


FIGURE 3.3.3 Lithium and sodium atoms both have a core charge of +1.

As for all group 1 elements, the valence electron of a lithium atom and a sodium atom experience a core charge of +1.

Moving down a group you can see that the core charge remains constant. Thus, any differences in the strength of the attraction between the valence electron in lithium and sodium, and their respective nuclei, cannot be explained in terms of differences in charge. However, the number of electron shells increases.

Hence, the valence electron(s) are further from the nucleus. This increase in distance has a marked effect on the force of attraction the valence electron(s) experience. Consequently, they will be pulled less strongly towards the nucleus.

Now consider sodium and chlorine. They are both in period 3 in the periodic table (Figure 3.3.4).

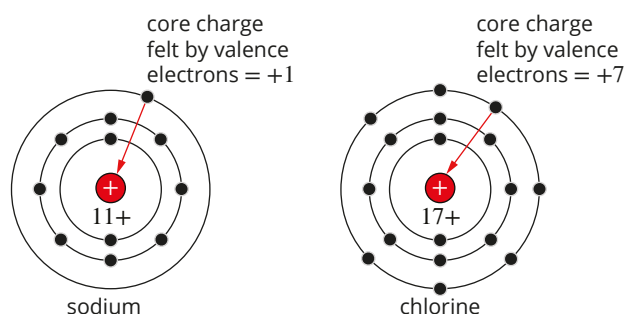


FIGURE 3.3.4 The core charges of two period 3 elements, sodium and chlorine, are +1 and +7 respectively.

Moving across a period you can see that the number of electron shells remains constant. Thus, any differences in the strength of the attraction between the valence electron(s) in sodium and chlorine, and their respective nuclei, cannot be explained in terms of differences in distance. However, the core charge increases.

Hence, the valence electron(s) are attracted by an increasing core charge. This increase in charge affects the force of attraction the valence electron(s) experience. Consequently, they will be pulled more strongly towards the nucleus.

Table 3.3.1 summarises how the attraction between the nucleus and valence electrons changes in the periodic table.

TABLE 3.3.1 The changes in attraction between the nucleus and valence electrons within groups and periods of the periodic table

	Trend in core charge	Trend in nuclei-valence electron distance	Trend in attraction between the nucleus and valence electrons
down a group	remains constant	increases	The valence electrons are more weakly held moving down a group as, although the core charge stays constant, the valence electrons are further from the nucleus (there are more shells in the atom) and thus experience a weaker electrostatic attraction.
left to right across a period	increases	remains constant	The valence electrons are more strongly held moving across a period as, although the distance between the valence electrons and the nucleus distance remains essentially constant (same electron shell), the valence electrons are subject to a higher core charge and thus experience a stronger electrostatic attraction.

The trends of atomic radii, ionisation energy and electronegativity are easily explained in terms of the strength of attraction experienced by the valence electrons.

ATOMIC RADIUS

Atomic radius is a measurement used for the size of atoms. It can be regarded as the distance from the nucleus to the valence shell electrons. Figure 3.3.5 depicts the atomic radii of many of the main group elements.

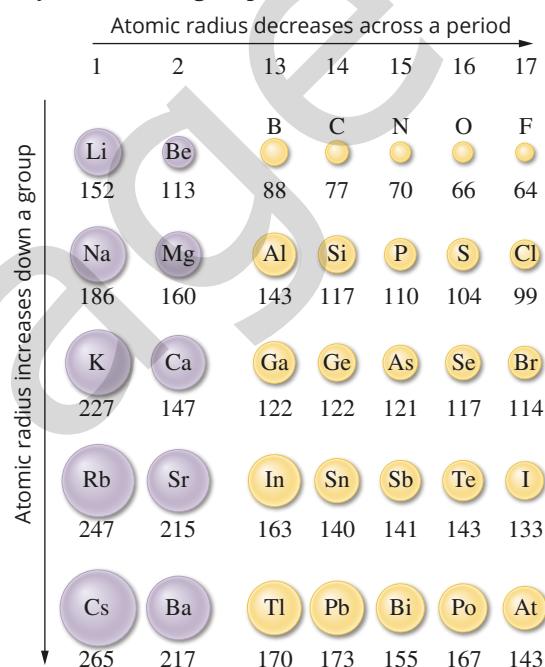


FIGURE 3.3.5 The relative sizes of atoms of selected main group elements. The atomic radii are given in picometres (pm). A picometre is 10^{-12} m.

Atoms do not have sharply defined boundaries and so it is not possible to measure their radii directly. One method of obtaining atomic radii, and therefore an indication of atomic size, is to measure the distance between nuclei of atoms of the same element in molecules. For example, in a hydrogen molecule (H_2) the two nuclei are 74 picometres (pm) apart. The radius of each hydrogen atom is assumed to be half of that distance, i.e. 37 pm.

Table 3.3.2 explains the trends in atomic radius in the periodic table.

TABLE 3.3.2 Trends in atomic radius in the periodic table

	Trend in atomic radius	Explanation
down a group	increases	Core charge stays constant and the number of shells increases as you move down a group. As a result, atomic radius increases down a group.
left to right across a period	decreases	As you move across a period, the number of occupied shells in the atoms remains constant but the core charge increases. The valence electrons become more strongly attracted to the nucleus, so atomic radius decreases across a period.

IONISATION ENERGY

The valence electrons are held to a nucleus due to the electrostatic attraction between the two. If an atom is given sufficient energy, this electrostatic attraction can be overcome and an electron can be completely removed from the atom. The process of removing an electron from an atom and forming an ion is called **ionisation**.

The valence electrons are removed first because they are the furthest electrons from the nucleus and are the least strongly held. When ionisation occurs, the atom will now have one less electron than the number of protons in the nucleus, and becomes a positively charged ion.

EXTENSION

Ionising radiation

Radiation is a term that can have negative connotations in wider society but, scientifically, radiation is simply the transmission of energy via waves or particles across space. Electromagnetic radiation is in the form of waves, whereas alpha and beta radiation, generated from nuclear decay, is in the form of particles.

Certain high energy forms of radiation are classified as ionising radiation. This is radiation which carries with it sufficient energy to remove electrons from atoms or molecules, thereby ionising them. Exposure to ionising radiation poses significant health risks as the molecules within cells, such as DNA, can be damaged by ionising radiation, possibly leading to cancer.

Of the electromagnetic spectrum, gamma rays, X-rays and some ultraviolet light is considered ionising. Other forms of electromagnetic radiation, such as microwaves used in cooking appliances and radio waves used in mobile phones, are not energetic enough to cause the ionisation of atoms or molecules. These forms of radiation can only cause electrons to become 'excited', as they move from lower to higher energy levels (shells).

First ionisation energy

The energy required to remove one electron from an atom of an element in the gas phase is called the **ionisation energy**. For example, the ionisation energy of sodium is 494 kJ per mole of sodium atoms.

The **first ionisation energy** is the energy required to remove the first valence electron from an atom in the gas phase. Figure 3.3.6 shows the first ionisation energies of most main group elements.

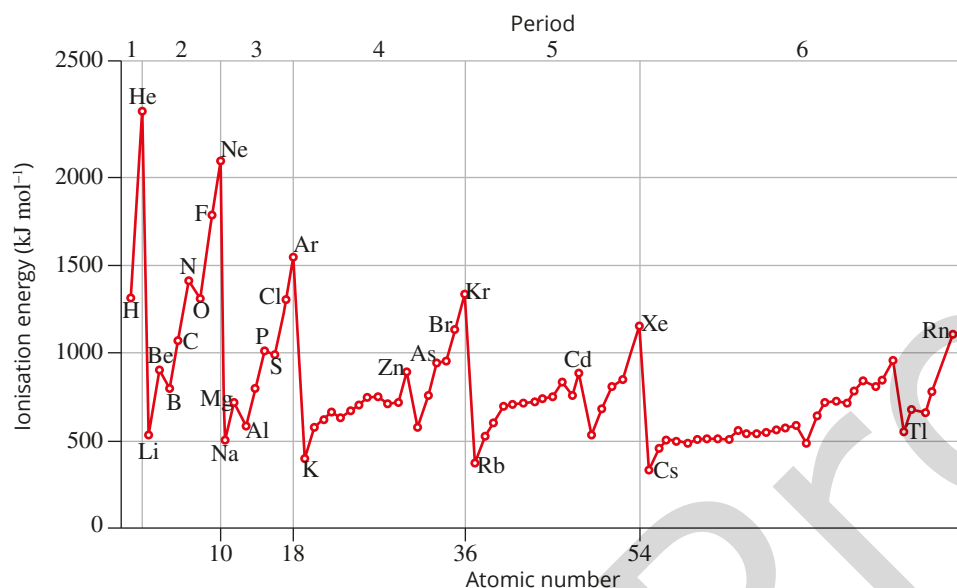


FIGURE 3.3.6 The first ionisation energy of a range of main group elements

The magnitude of the first ionisation energy reflects how strongly the valence electron is attracted to the nucleus of the atom. The more strongly the valence electron is attracted to the nucleus, the more energy is required to remove it from the atom and the higher the first ionisation energy.

The main group elements (groups 1, 2, 13–18) of the periodic table follow these general patterns. As the core charge increases across a period, so too does the first ionisation energy. As atomic radius increases down a group (due to the additional electron shell), the electrons are further from the nucleus. Therefore, the valence electrons in elements lower in a group can be removed more easily, meaning the first ionisation energy decreases.

TABLE 3.3.3 The trend in first ionisation properties in groups and periods of the periodic table

	Trend in ionisation energy	Explanation
down a group	decreases	Core charge stays constant and the number of shells increases as you move down a group. Therefore, the valence electrons are less attracted to the nucleus as they are further from the nucleus. As a result, the energy required to overcome the attraction between the nucleus and the valence electron is less, and the first ionisation energy decreases down a group.
left to right across a period	increases	Core charge increases and the number of occupied shells remains constant as you move across a period. As a result, the valence electrons become more strongly attracted to the nucleus, and more energy is required to remove an electron. Therefore, the first ionisation energy increases across a period.

Successive ionisation energy

After the removal of the first valence electron through ionisation, it is possible to then remove the remaining valence electrons and also the electrons which occupy the inner shells if sufficient energy is supplied. The energy required to achieve the sequential removal of electrons from the atom is called the **successive ionisation energy**.

Consider a sodium atom with a total of 11 electrons. As each successive electron is removed, these ionisation energies can be measured. Figure 3.3.7 shows that the first electron to be removed has the lowest ionisation energy and is therefore the easiest to remove. The following eight electrons are slightly more difficult, and finally, the last two require substantially more energy to remove.

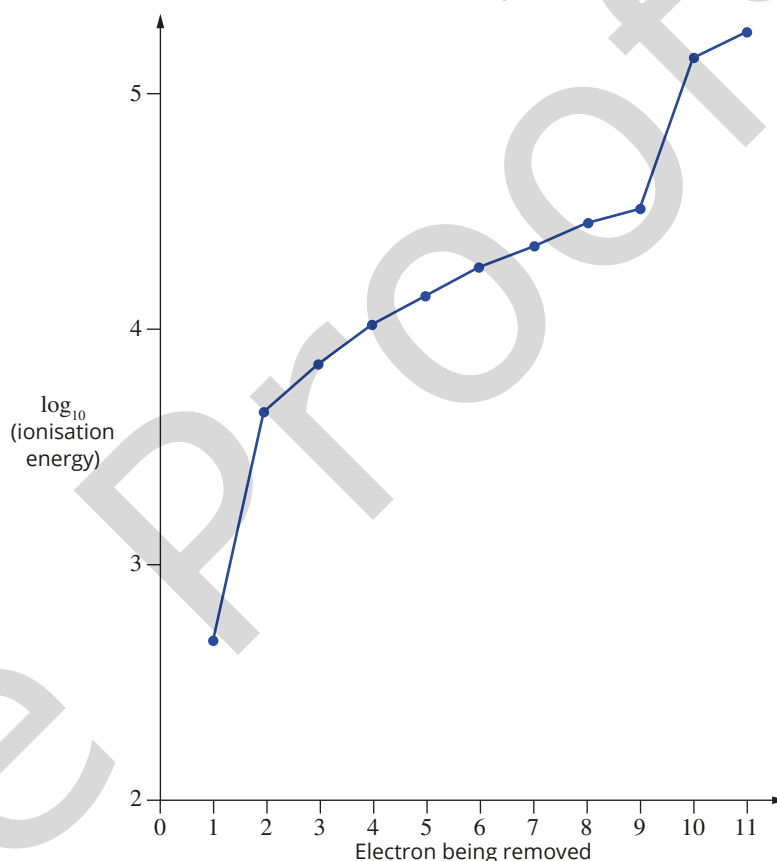


FIGURE 3.3.7 Graph of the ionisation energies of a sodium atom. (The logarithm of ionisation energy is used to provide a more convenient vertical scale.)

A sodium atom has the electron configuration of 2,8,1, which means there is only one valence electron in the third shell. As this electron is further away from the positive nucleus, and the electrostatic attraction to the nucleus it experiences is reduced by the shielding provided by the filled inner two shells of electrons, it is relatively weakly held in the atom compared to the other electrons and thus has the smallest ionisation energy.

The following eight electrons are all removed from the second shell of the sodium atom. Notice that there is a significant increase in ionisation energy from the first valence electron to the next eight electrons. As these eight electrons occupy the second shell, which is closer to the nucleus, they experience a greater electrostatic attraction than the first valence electron in the third shell and thus require a greater amount of energy to overcome this stronger attraction. Remember that, according to Coulomb's law, distance has a significant effect on the force of attraction between two charged particles. These electrons also receive less shielding from filled inner shells than the original valence electron in the third shell. This pattern is continued for the two electrons in the innermost shell. Being closest to the nucleus, they are the most tightly held electrons, requiring a much greater quantity of energy to remove them from the atom.

There is a gradual increase in the energy required to remove the second through to the eighth electrons. As these electrons all occupy the second shell, they all exist at similar distances from the nucleus, so the increasing energy requirement to remove them cannot be attributed to the changes in the distance to the nucleus. Each successive removal of an electron does, however, result in an increasingly positively charged ion, as the protons in the nucleus no longer have sufficient electrons to balance their positive charge.

The removal of subsequent electrons from an increasingly positively charged ion (Figure 3.3.4) will require a greater quantity of energy to overcome the greater electrostatic attraction experienced by each electron (Table 3.3.4).

TABLE 3.3.4 The ionisation equations for the successive ionisations of sodium

Electron being removed	Ionisation equation	Ionisation energy (kJ mol ⁻¹)
1	$\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$	496
2	$\text{Na}^+ \rightarrow \text{Na}^{2+} + \text{e}^-$	4562
3	$\text{Na}^{2+} \rightarrow \text{Na}^{3+} + \text{e}^-$	6910
4	$\text{Na}^{3+} \rightarrow \text{Na}^{4+} + \text{e}^-$	9543
5	$\text{Na}^{4+} \rightarrow \text{Na}^{5+} + \text{e}^-$	13 354
6	$\text{Na}^{5+} \rightarrow \text{Na}^{6+} + \text{e}^-$	16 613
7	$\text{Na}^{6+} \rightarrow \text{Na}^{7+} + \text{e}^-$	20 117
8	$\text{Na}^{7+} \rightarrow \text{Na}^{8+} + \text{e}^-$	25 496
9	$\text{Na}^{8+} \rightarrow \text{Na}^{9+} + \text{e}^-$	28 932
10	$\text{Na}^{9+} \rightarrow \text{Na}^{10+} + \text{e}^-$	141 362
11	$\text{Na}^{10+} \rightarrow \text{Na}^{11+} + \text{e}^-$	159 076

i Ionisation energy is the energy required to completely remove an electron from an atom. This energy increases as more electrons are removed from the atom.

METALLIC CHARACTER

Metals conduct electricity and are usually solids at room temperature. Conversely, non-metallic elements usually do not conduct electricity and many are gases at room temperature.

The differences between the properties of metals and non-metals are related to the number of electrons in the outer shell of their atoms. In general, elements with atoms containing one, two or three valence electrons tend to behave as metals, whereas those with four or more valence electrons behave as non-metals.

Metals are located on the left side of the periodic table and non-metals are on the right. Elements known as the **metalloids** are located between the metals and non-metals. The metalloids exhibit both metallic and non-metallic properties. Silicon is one of the most abundant metalloids. It is a brittle solid, which is a common property of non-metals. However, it is also a semiconductor, meaning it can exhibit electrical conductivity, making it useful in many electronic devices such as computers and calculators.

The classification of an element as metallic or non-metallic, however, is not always clear. Across the periodic table, the change from metals to non-metals can be considered as a continuum. **Metallic character** is the name given to describe how closely an element exhibits the properties commonly associated with metals, namely, that it readily loses an electron to form a cation. This is closely related to ionisation energy.

As seen previously, the first ionisation energy decreases as you move down a group due to the increasing distance to the nucleus (due to additional electron shells) and from the right to left of a period (due to the reducing core charge). The elements with the most weakly held valence electrons, having the smallest first ionisation energy and the greatest metallic character occur towards the bottom left-hand side of the periodic table.

ELECTRONEGATIVITY

Electronegativity is the ability of an atom to attract electrons in a covalent bond towards itself. The more strongly the valence electrons of an atom are attracted to the nucleus of the atom, the greater the electronegativity. The strength of this attraction, as for all electrostatic depends on the magnitude of the charges involved and the distance between them. Figure 3.3.8 shows the electronegativity of many of the main group elements.

Electronegativity increases across a period. →

	1	2		13	14	15	16	17
	Li	Be		B	C	N	O	F
	1.0	1.6		2.0	2.6	3.0	3.4	4.0
	Na	Mg		Al	Si	P	S	Cl
	0.9	1.3		1.6	1.9	2.2	2.6	3.2
	K	Ca		Ga	Ge	As	Se	Br
	0.8	1.0		1.8	2.0	2.2	2.6	3.0
	Rb	Sr		In	Sn	Sb	Te	I
	0.8	1.0		1.8	2.0	2.1	2.1	2.7
	Cs	Ba		Tl	Pb	Bi	Po	At
	0.8	0.9		2.0	2.3	2.0	2.0	2.2
	Fr	Ra						
	0.7	0.9						

Electronegativity decreases down a group.

FIGURE 3.3.8 The electronegativity of elements generally decreases down a group and increases across a period, from left to right.

The values given for electronegativity are dimensionless quantities, that is, they have no unit and are calculated theoretically rather than measured experimentally.

The trends observed in the electronegativity of the elements are summarised in Table 3.3.5.

TABLE 3.3.5 Trends in electronegativity in groups and periods of the periodic table

	Trend in electronegativity	Explanation
down a group	decreases	The core charge stays constant and the number of shells increases down a group. Therefore, valence electrons are less strongly attracted to the nucleus as they are further from the nucleus. As a result, electronegativity decreases.
left to right across a period	increases	The number of occupied shells in the atoms remains constant but the core charge increases across a period. Therefore, the valence electrons become more strongly attracted to the nucleus. As a result, electronegativity increases.

Although electronegativity is related to the attraction of electrons within a covalent bond, the electronegativity values are useful in predicting the types of bonding in a chemical and are therefore an important concept in Chemistry. This will be covered in greater depth in Chapters 5 and 6.

3.3 Review

SUMMARY

- The attractive force between opposite electric charges is known as electrostatic attraction.
- The strength of an electrostatic attraction is dependent on the magnitude of the charges involved and the distance between them.
- Core charge or effective nuclear charge is the resultant attractive force experienced by valence electrons once the impact of the shielding effect provided by electrons in inner shells is taken into account.
- The core charge or effective nuclear charge is calculated by subtracting the total number of inner-shell electrons from the number of protons in the nucleus.
- Atomic radius is a measurement used for the size of atoms. It can be regarded as the distance from the nucleus to the outermost electrons.
- The first ionisation energy is the energy required to remove the first valence electron from an atom of an element in the gas phase.
- First ionisation energy decreases down a group but increases across a period.
- Successive ionisation energies are the energies required to remove multiple electrons consecutively from an atom of an element in the gas phase.
- Electronegativity is the ability of an element to attract electrons in a covalent bond towards itself.
- Table 3.3.6 summarises how certain properties have specific trends within the groups and periods of the periodic table.

TABLE 3.3.6 Property trends in the periodic table

Property	Down a group	Across a period (left to right)
core charge	no change	increases
atomic radius	increases	decreases
ionisation energy	decreases	increases
electronegativity	decreases	increases

- Many trends in the physical properties of elements in the periodic table can be explained using two key ideas.
 - From left to right across a period, the core charge of atoms increases, so the attractive force felt between the valence electrons and the nucleus increases.
 - Down a group, the number of shells in an atom increases so that the valence electrons are further from the nucleus and are held less strongly.

KEY QUESTIONS

- 1 What is the core charge of an atom of carbon?
- 2 Explain why atomic radius decreases as you move left to right across a period, yet the number of protons and neutrons in the nucleus increases.
- 3
 - a Explain the meaning of the term 'ionisation energy'.
 - b What factors need to be considered when predicting the trend in first ionisation energy across a period?
- 4 Explain why ionisation energy increases from left to right across a period.
- 5 Figure 3.3.8 gives electronegativity values for the elements in groups 1, 2 and 13–17 of the periodic table.
 - a Give the name and symbol of the element that has the:
 - i highest electronegativity
 - ii lowest electronegativity.
 - b In which group do you see the:
 - i greatest change in electronegativity as you go down the group?
 - ii smallest change in electronegativity as you go down the group?
 - c Why are the elements of group 18 usually omitted from tables that give electronegativity values?
- 6 Explain the relationship between electronegativity and core charge.

3.4 Quantisation of energy

The fact that metals, when heated strongly, emit light of a specific colour which is characteristic of the given element was a well-known phenomenon throughout the early 1800s. In fact, scientists were using this practice to confirm the presence of certain metals in substances long before Bohr proposed a model of the atom which enabled its explanation.

QUANTISED ENERGY OF ELECTRONS

What allowed Bohr's model to explain these light emissions was to identify that the electrons in atoms can only be found at certain discrete energies. These are the energies associated with the shells in which the electrons can be found. The energy levels of the electrons are described as being **quantised**.

Bohr's model allowed electrons to jump between these discrete energy levels or shells. They accomplish this by either absorbing or emitting energy.

Consider a hydrogen atom with one proton and one electron. Usually, the electron exists in the $n = 1$ shell. This is the lowest energy state of the atom and is called the **ground state**.

When the hydrogen atom is heated strongly the electron absorbs the energy and jumps to a higher energy level. This higher energy state is known as the **excited state**.

The higher energy excited state is inherently unstable and remains so for only fractions of a second. The excited atom then emits the recently absorbed energy as the electron returns to a lower energy level. The energy which is emitted by the atom is in the form of electromagnetic radiation.

Electromagnetic radiation

Visible light is a form of energy and a type of **electromagnetic radiation**. Other forms of electromagnetic radiation include radio waves and X-rays. The range of different forms of electromagnetic radiation is referred to as the **electromagnetic spectrum**, as seen in Figure 3.4.1.

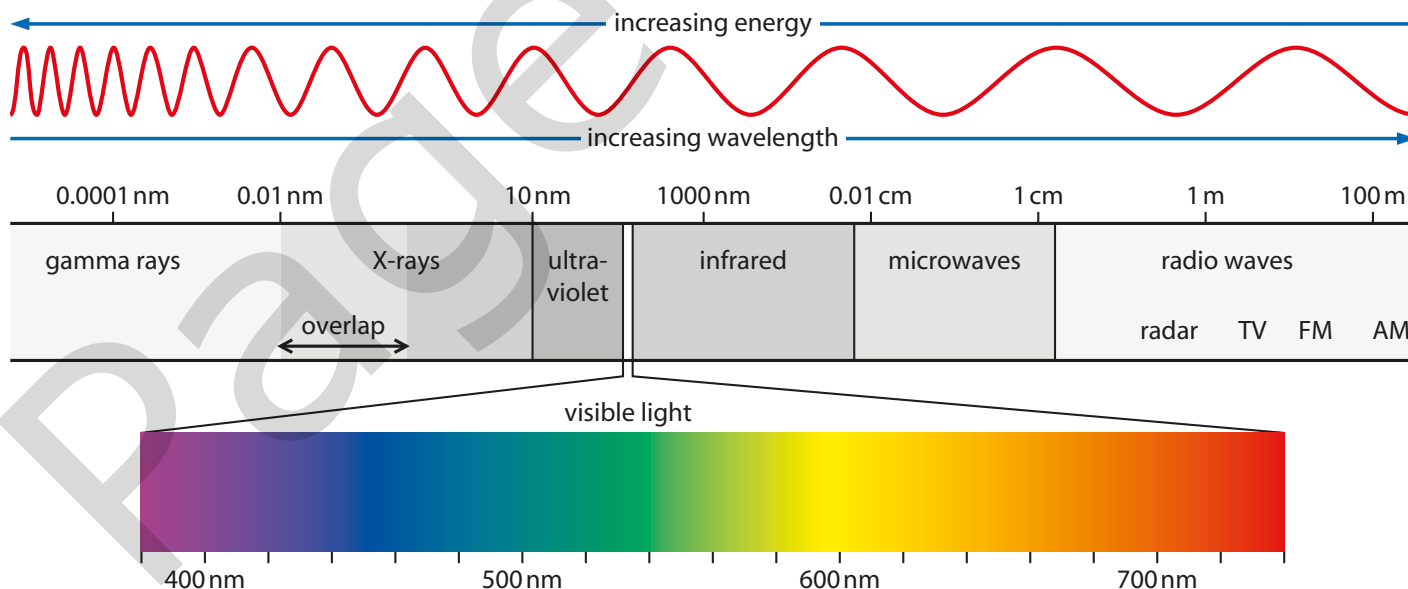


FIGURE 3.4.1 Visible light is only a small part of the electromagnetic spectrum.

Radiation from each portion of the electromagnetic spectrum can be described in terms of its frequency, wavelength and energy. Different colours of light can also be described in terms of different frequencies, wavelengths and energies. For example, compared to red light, violet light has a higher energy and higher frequency, but a shorter wavelength.

While excited atoms returning to their ground state emit electromagnetic radiation, atoms in their ground state can also absorb certain amounts of electromagnetic radiation.

By observing and measuring either the electromagnetic radiation absorbed by atoms in their ground state or that which is emitted by excited atoms, scientists can build up a picture of the energy levels present in the atom. As the arrangement and interactions of the electric charges in an atom are unique to that particular element, so is the energy of the absorbed or emitted electromagnetic radiation. This electromagnetic radiation can be observed as either absorption spectra or emission spectra.

CHEMFILE

Uranium glass

Uranium is a heavy, dense, grey metal which is commonly used as a fuel for nuclear power due to its radioactive U-238 isotope.

From around the late 1800s to early 1900s uranium oxide was popular as a glass colourant, producing a range of yellow to green coloured decorative glass. Most pieces usually only contain trace levels of uranium, around 2%, but some pieces contain up to 25% uranium. Uranium glass is slightly radioactive but is still considered safe.

A peculiar property of uranium glass is it fluoresces bright green when exposed to ultraviolet light (Figure 3.4.2). The process of fluorescence is similar to that of atoms emitting radiation as their electrons return from an excited state to a lower energy state, except that in the case of fluorescence the emitted radiation is of a different colour from the incident radiation.

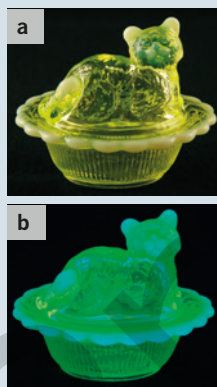


FIGURE 3.4.2 A piece of uranium glass viewed (a) under visible light and (b) under ultraviolet light

ABSORPTION SPECTRA

Electrons in energy levels close to the nucleus have the lowest energies and experience the strongest attraction to the nucleus. An electron can jump up to a higher energy level if it absorbs energy that corresponds exactly to the difference in energy between the lower energy level and the higher energy level (Figure 3.4.3).

When an electron jumps to a higher energy level, it is not necessarily the level directly above it. For example, in the case of a hydrogen atom with a single valence electron, if it is exposed to sufficient energy, the valence electron can jump to the second, third or even fourth energy level. As these energy levels represent distinct, quantised energies, the energy required to promote a ground state electron to these higher levels are all different, but identical for any hydrogen atom.

The different quantities of energy required to promote these ground state electrons are supplied by electromagnetic radiation of varying wavelength and frequency, which are observed as different colours of light (Figure 3.4.4).

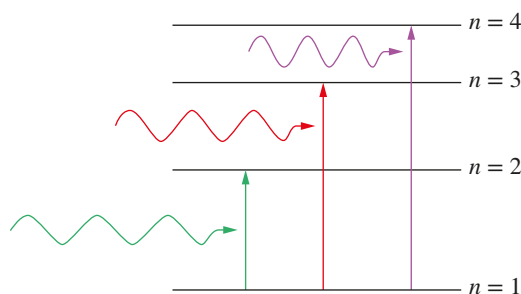


FIGURE 3.4.4 Energy of a specific wavelength and frequency corresponds to jumps between specific energy levels (electron shells) in the atom.

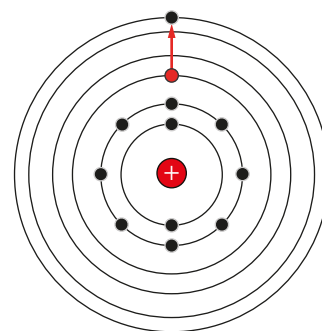


FIGURE 3.4.3 Absorbing energy can cause an electron to jump to a higher energy level.



FIGURE 3.4.5 The absorption spectrum of hydrogen

i The black lines in an absorption spectrum represent electrons gaining energy as they move to higher energy levels in the atom.

A sample of hydrogen atoms in their ground state exposed to a continuous spectrum of visible light consisting of every possible wavelength will only absorb certain colours. These colours have energies that perfectly correspond to the energy needed to promote an electron to a higher energy level. The individual colours of light in this continuous spectrum that are absorbed by the hydrogen atoms are known as **absorption lines**. Collectively, these absorption lines form an **absorption spectrum**. The absorption spectrum for hydrogen is shown in Figure 3.4.5.

The dark regions in the spectrum correspond to the colours of light that provide the exact quantity of energy required to promote its valence electrons to a higher energy level. Even though there is only one electron in each hydrogen atom, there are several absorptions lines because each individual line is caused by the promotion of an electron to the second, third or fourth energy level or higher. In any sample of hydrogen there will be many hydrogen atoms and this absorption spectrum represents the collective absorptions of all individual hydrogen atoms.

CHEMISTRY IN ACTION

Discovering the chemical composition of the Sun and the planets

In 1814, Joseph von Fraunhofer, a German physicist and skilled glassmaker, noticed a series of dark regions in the visible spectrum of the Sun. He made a number of systematic measurements of these lines; in total, cataloguing over 570 distinct lines.

About 45 years later Gustav Kirchhoff, another German physicist and Robert Bunsen, a German chemist, noticed that the lines Fraunhofer documented coincided with the characteristic emission lines of a number of elements. They correctly deduced that the dark regions are absorption lines of the chemical elements in the Sun. Thus they were able to determine the chemical composition of the Sun. These absorption lines are known as Fraunhofer lines (Figure 3.4.6) in his honour.

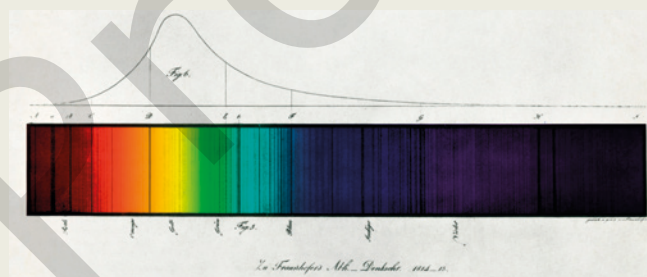


FIGURE 3.4.6 Fraunhofer's original documentation of the Sun's absorption lines

These principles are still employed today in astronomical spectroscopy where astronomers measure the electromagnetic spectra emitted by stars and other hot celestial bodies to determine their chemical composition.

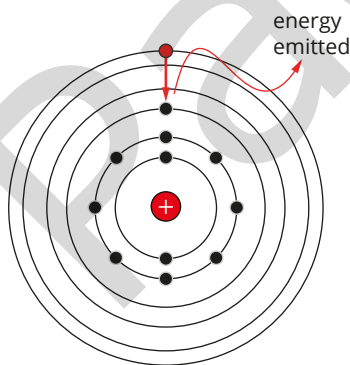


FIGURE 3.4.7 An excited electron quickly returns to a lower energy level, emitting electromagnetic energy such as light of a particular energy.

EMISSION SPECTRA

Heating an element vigorously can cause an electron to absorb energy and jump to a higher energy state. Shortly afterwards, the electron returns to a lower energy state. As the electron falls to a lower energy shell, it emits energy in the form of light that corresponds exactly to the difference in energy between the higher and the lower energy levels (Figure 3.4.7).

The electron can return to its original ground state in a number of different pathways. The electron may return directly to the ground state or may move to other energy levels before returning to the ground state (Figure 3.4.8). For example, an electron in the $n = 4$ shell may move to the $n = 2$ shell before returning to the $n = 1$ ground state.

As these energy levels represent distinct, quantised energies, each transition the electron makes is associated with the emission of a different quantity of energy in the form of different colours of light. These colours will be consistent for all hydrogen atoms.

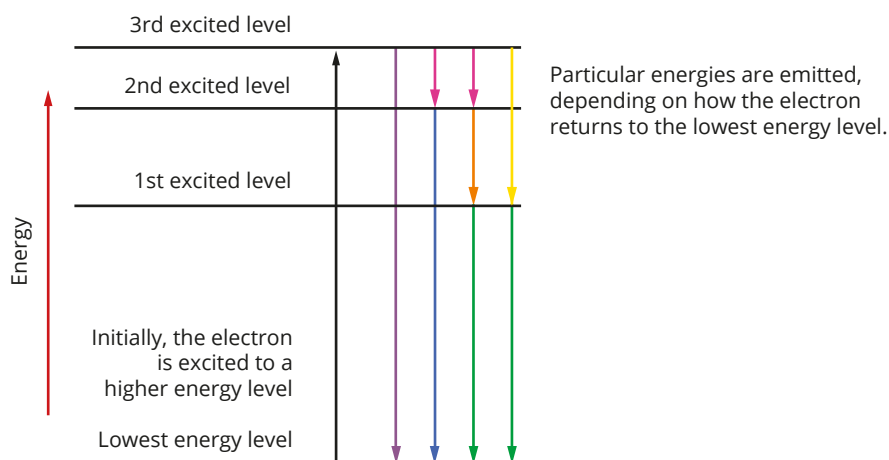


FIGURE 3.4.8 Emission of energy as electrons move from a higher to a lower energy state

So if a sample of hydrogen atoms is heated to an excited state and its valence electron jumps to some higher energy level, moments later as it returns to a lower energy level it will emit a colour of light whose energy perfectly corresponds to the energy difference between the two levels. The individual colours of light that are emitted by the hydrogen atoms are known as **emission lines** and collectively these emission lines form an **emission spectrum**. The emission spectrum for hydrogen is shown in Figure 3.4.9.

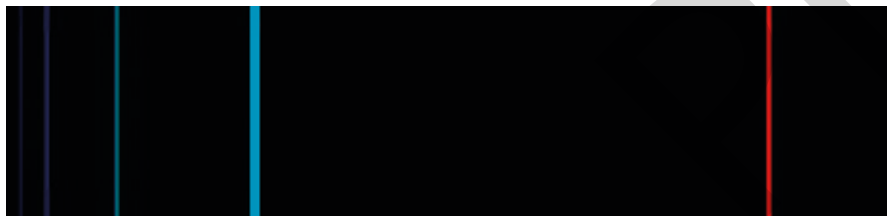


FIGURE 3.4.9 The emission spectrum of hydrogen

Even though a hydrogen atom has only a single electron, there are several lines in the spectrum because electrons can jump between different energy levels. As is the case in absorption spectra, this emission spectrum represents the collective emissions of many individual hydrogen atoms.

i The coloured lines in an emission spectrum represent electrons losing energy as they move back down to lower energy levels in the atom.

CHEMFILE

How helium got its name

Helium was discovered on the Sun before being discovered on Earth. The French astronomer Jules Janssen discovered helium in 1868 while studying the light from a solar eclipse in India similar to the solar eclipse shown in Figure 3.4.10.

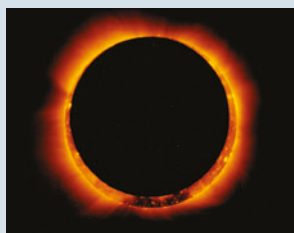


FIGURE 3.4.10 The emission spectrum of helium was first detected in the sunlight of a solar eclipse like the one shown here.

Although the spectrum showed the full range of colours, the bright yellow line in the helium spectrum (Figure 3.4.11) stood out. The line could not be matched to the line spectra of any of the known elements. It was concluded that the line belonged to an unknown element. This element was named helium after the Greek sun god, Helios.



3.4.11 The emission spectrum of helium is made up of lines ranging from violet to red in colour.

3.4 Review

SUMMARY

- The Bohr model of the atom was the first atomic model to explain the origin of emission spectra.
- Bohr's model of the atom introduced the quantisation of energy, that is, the energy of the electrons only exist at certain discrete levels.
- The lowest or minimum energy state of an atom is called the ground state.
- An elevated or higher energy state an atom can find itself in is called an excited state.
- When an electron absorbs energy (e.g. heat or light), the electron can jump from one energy level (shell) to a higher energy level.
- Radiation from each portion of the electromagnetic spectrum can be described in terms of its frequency, wavelength and energy. Different colours of light have different frequencies, wavelengths and energies.
- The range of colours of light absorbed by electrons moving into higher energy levels are called absorption spectra.
- When an electron falls from a higher energy level to a lower energy level, it emits energy in the form of light.
- The range of colours of light emitted by electrons moving into lower energy levels is called emission spectra.
- Each element has a unique absorption/emission spectrum.
- Each line in the absorption/emission spectrum corresponds to a specific electron transition between two energy levels.

KEY QUESTIONS

- 1 Explain how emission spectra provide evidence for electron shells in the Bohr model.
- 2 Explain why the lines in an emission spectrum are different colours.
- 3 Which of the following represents the electron configuration of a carbon atom in its ground state?
A 2,4
B 2,5
C 2,3,1
D 2,4,2
- 4 Which of the following represents the electron configuration of a carbon atom in its excited state?
A 2,4
B 2,5
C 2,3,1
D 2,4,2
- 5 Does a sodium ion (2,8) and a neon atom (2,8) display the same emission spectra? Explain your answer.
- 6 Do all isotopes of an element display the same emission spectra? Explain your answer.

3.5 Spectroscopy

As each element has a different number of protons in the nucleus, the attraction of the nucleus for electrons will vary from element to element. No two elements therefore have energy levels of exactly the same energy, so an emission or absorption spectrum is characteristic of a particular element. It may be used as a 'fingerprint' to identify the elements present in a substance.

The study of the interaction between matter and electromagnetic radiation is known as spectroscopy and is the basis for a number of qualitative and quantitative analytical techniques.

FLAME TESTS

Chemists use the fact that some metals produce particular colours when they are heated as a convenient and simple method of analysis. These different colours are caused by the different electron transitions occurring in the different atoms. The metallic elements present in a compound can often be determined just by inserting a sample of the compound into a non-luminous Bunsen burner flame in a test known as a **flame test** (Figure 3.5.1).

The metal is identified by comparing the flame colour with the characteristic colours produced by metals. Some examples of the flame colours can be seen in Figure 3.5.2.

Table 3.5.1 below shows some of the flame colours that can be used to identify the presence of metals using a flame test. It is important to note that only a small number of metals produce flame colours as the emission lines of some elements are not within the visible spectrum of light.

TABLE 3.5.1 Common metal flame test colours

Metal	Flame colour
sodium	yellow
strontium	scarlet
copper	green
barium	yellow-green
lithium	crimson
calcium	red
potassium	lilac

While the flame test provides a quick analysis of the chemical composition of a sample, there are a number of limitations in this technique.

- The flame test is a qualitative test only.
- Only a small range of metals are detectable by the flame test.
- Metals in low concentrations may be difficult to observe.
- Mixtures of metals will produce confusing results.

ATOMIC ABSORPTION SPECTROSCOPY

There is a direct mathematical relationship between the concentration of the sample being analysed in a sample and the intensity of light it absorbs. This relationship is used in the technique of **atomic absorption spectroscopy** (AAS).

In AAS, the sample being analysed is vaporised in a flame to produce free atoms in their ground state. This cloud of ground state atoms then passes through the optical path of a light source which produces the emission spectrum of the particular chemical species being examined.

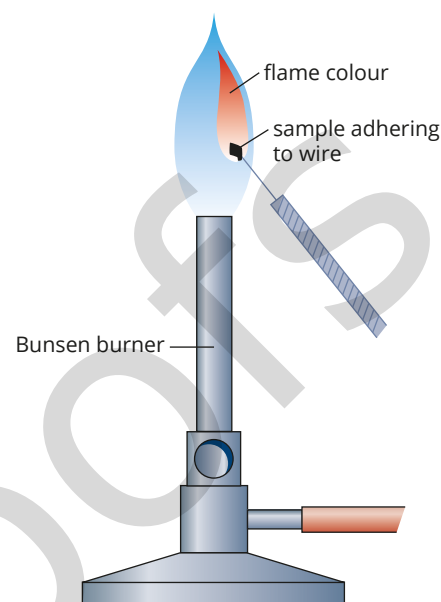


FIGURE 3.5.1 Performing a flame test. A moist wire has been dipped in the sample and then placed in the flame. A fine spray of solution from a spray bottle could be used instead.



FIGURE 3.5.2 The colour of a flame is different in the presence of different metal compounds and can be used to identify the presence of these metals. The flame colours shown here are for (left to right): lithium (red), strontium (red), sodium (yellow), copper (green) and potassium (lilac).

As the light source produces light of the same energy required to raise ground state atoms to an excited state, the cloud of gaseous atoms absorb some portion of the incident light.

The intensity of the light which is absorbed is measured, giving an indication of the concentration in the original sample.

Atomic absorption spectroscopy is employed by many modern-day chemists (Figure 3.5.3) and addresses some of the limitations of flame tests.

- AAS provides both a qualitative and quantitative test.
- More than 70 elements can be analysed by AAS.
- AAS can detect elements in concentrations as low as micrograms per litre.
- AAS is highly selective and is regularly used with mixtures.

i Flame tests and atomic absorption spectroscopy can be used to identify the presence of certain atoms in a chemical sample.



FIGURE 3.5.3 An analytical chemist uses AAS to determine the concentration of metals in food.

CHEMFILE

Sir Alan Walsh

The modern form of AAS was largely developed in the early 1950s by a team of Australian researchers at the CSIRO lead by Sir Alan Walsh.

At the time, the main analysis technique focused on the emission of characteristic radiation, rather than the absorption of it. After a 'eureka' moment in his garden, he constructed a working prototype by the very next day. Over the next few years, Walsh and his researchers released a number of publications detailing their new technique. Unfortunately, there was very little interest from other scientists. Walsh's reaction, as documented in a 1970 interview was 'It was only because, really, of the cold shoulder treatment we were given that we got a bit mad and got involved in proving a point. So if the world had reacted differently, we might have come out of this with nothing'.

Alan Walsh (Figure 3.5.4) was knighted in 1977 for 'services to science', the year of his retirement from the CSIRO in Melbourne.



FIGURE 3.5.4 Sir Alan Walsh, the inventor of atomic absorption spectroscopy

3.5 Review

SUMMARY

- Flame tests are based on the ability of electrons to release absorbed energy as they move from higher to lower energy levels.
- Flame tests are performed by inserting a sample in a non-luminous Bunsen burner flame.
- Flame tests can be used to detect the presence of a small number of metal elements.
- Atomic absorption spectroscopy (AAS) is based on the ability of electrons to absorb energy as they move from lower to higher energy levels.
- AAS can be used to detect a large range of elements, even at low concentrations a sample.

KEY QUESTIONS

- 1 What is spectroscopy?
- 2 Describe the advantages and disadvantages of a flame test.
- 3 In atomic absorption spectroscopy, what is the source of the energy that excites the atoms in the sample being analysed?
- 4 What are some of the advantages of atomic absorption spectroscopy over a flame test?
- 5 Flares are small pyrotechnic devices used for signalling or illumination. Red flares are traditionally used in maritime environments to indicate distress. Use Table 3.5.1 on page xx to suggest which metal is likely present in the flare's chemical composition.

Chapter review

KEY TERMS

absorption line	electromagnetic spectrum	inert	
absorption spectrum	electron configuration	ion	
anion	electron shell	ionisation	
atomic absorption	electronegativity	ionisation energy	successive ionisation energy
spectroscopy (AAS)	emission line	main group element	transition metal
atomic radius	emission spectrum	metallic character	valence electron
Bohr diagram	energy level	metalloid	valence shell
Bohr model	excited state	noble gas	
cation	first ionisation energy	octet rule	
core charge	flame test	period	
effective nuclear charge	ground state	quantised	
electromagnetic radiation	group		

Electronic structure of atoms

- 1 What is the fundamental difference between the planetary model of the atom put forward by Rutherford and Bohr's model of the atom?
- 2 Bohr's model of the atom describes an atom as consisting of rapidly moving electrons at a relatively large distance from a very small central nucleus. What is there *between* those electrons and the nucleus?
- c What is the name, symbol and electron configuration of the:
 - i second element in group 14?
 - ii second element in period 2?
 - iii element that is in group 18 and period 3?
- 4 Determine the period and group of the elements with the following electron configurations.
 - a 2,2
 - b 2,7
 - c 2,8,4
 - d 2,8,8,2
- 5 Name an element with properties similar to those of:
 - a carbon
 - b magnesium
 - c iodine
 - d phosphorus.
- 6 Main group elements are elements that belong to specific groups in the periodic table. Which groups are these?
- 7 How many valence electrons are in atoms of elements found in:
 - a group 1?
 - b group 15?
 - c group 17?
 - d group 2?

Electron arrangement in the periodic table

- 3 Use the periodic table in Figure 3.2.2 on page xx to answer these questions.
 - a In which groups of the periodic table are the following found?
 - i B
 - ii Cl
 - iii Na
 - iv Ar
 - v Si
 - vi Pb
 - b In which periods of the periodic table are the following found?
 - i K
 - ii F
 - iii He
 - iv H
 - v U
 - vi P

Trends in the periodic table

- 8** Across a period, the number of subatomic particles in an atom increases but the size of an atom decreases. Why?
- 9** Account for the fact that it takes more energy to remove an electron from the outer shell of atoms of:
- a** phosphorus than magnesium
 - b** fluorine than iodine.
- 10** Consider the elements in period 2 of the periodic table: lithium, beryllium, boron, carbon, nitrogen, oxygen, fluorine. Describe the changes that occur across the period. Consider:
- a** the sizes of atoms
 - b** metallic character
 - c** electronegativity.
- 11** From each set of elements, select the element that has the largest first ionisation energy.
- a** phosphorus, arsenic, nitrogen
 - b** silicon, chlorine, sulfur
 - c** bromine, chlorine, sulfur
- 12** Explain which of oxygen, chlorine and fluorine will have the largest first ionisation energy.
- 13** Why is the ionisation energy of an atom always measured for gaseous atoms?
- 14** When a sodium atom forms a cation, is the radius of the resultant ion larger or smaller than the original atom? Explain the reasoning for your answer.
- 15** When a fluorine atom forms an anion, the radius of the ion is larger than the original atom. Suggest a reason why this is so.

Quantisation of energy

- 16** In a hydrogen atom, which electron shell will the electron be in if the atom is in the ground state?
- 17** For any given atom, which is likely to require the greater amount of energy and why; the excitation of the atom or the ionisation of the atom?
- 18** Suggest a reason for the slightly blue colour present in a non-luminous Bunsen burner flame.
- 19** When Kirchhoff and Bunsen reviewed Joseph von Fraunhofer's catalogue of absorption lines in sunlight, they found absorption lines characteristic of oxygen. Considering there is no oxygen in outer space and very little in the Sun, explain the likely cause of these absorption lines.
- 20** Why does an emission spectrum contain a number of lines of different colours?

Spectroscopy

- 21** Why is it important to conduct a flame test in a non-luminous flame?
- 22** What would be the easiest way to produce the characteristic radiation required for atomic absorption spectroscopy?
- 23** Why would a sample containing a number of different metals not be suitable for analysis by flame test but possibly acceptable for analysis by atomic absorption spectroscopy?

Connecting the main ideas

- 24** New models for the atom evolve as scientists become aware of inconsistencies between current models and experimental data. Outline the problems with the existing model of the atom that led to the modifications suggested by the following scientists.
- a** Rutherford
 - b** Bohr

CHAPTER REVIEW CONTINUED

- 25 What concepts or ideas are the two versions of the periodic table below trying to convey?

